



Mini Project 01 Report

Two-Stage Miller OTA Design

Analog IC Design | Cadence Virtuoso

Student: Ahmed Ibrahim Ashry Mohammed **Prepared by:** Dr. Hesham Omran
Wave: su2026 ITI AICs **Supervised by:** Eng. Abdelaziz Mohammed
 Eng. Abdalrhman Alkafass

Part 1: g_m/I_D Design Charts

Question

Using ADT Device Xplore, plot the following design charts vs g_m/I_D for both PMOS and NMOS: g_m/g_{ds} , I_D/W , g_m/C_{gg} , and V_{GS} . Set $V_{DS} = V_{DD}/3$ and sweep $L = 0.18\mu, 0.5\mu : 0.5\mu : 2\mu$.

Answer:

I used the ADT Sizing Assistant to get the lookup table plots for both NMOS and PMOS transistors. The charts below show all the required parameters swept across g_m/I_D .

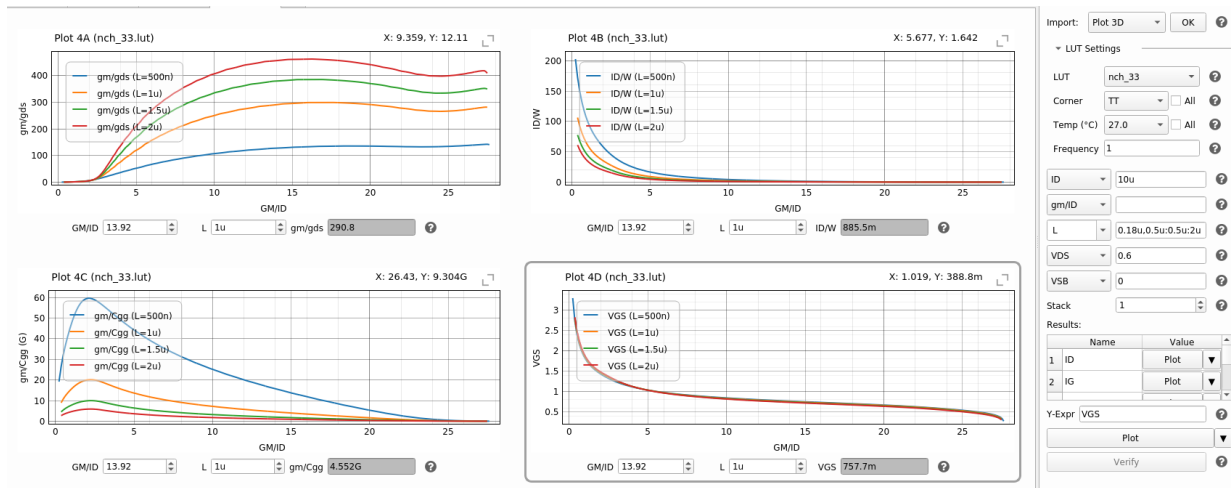


Figure 1. NMOS ADT Charts for g_m/g_{ds} , I_D/W , g_m/C_{gg} , and V_{GS} .

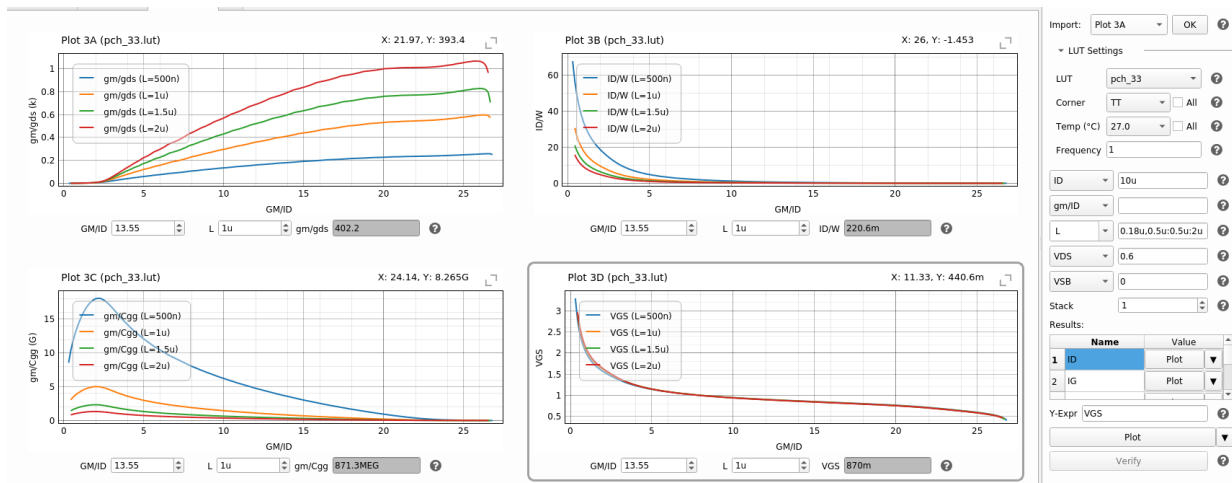


Figure 2. PMOS ADT Charts for g_m/g_{ds} , I_D/W , g_m/C_{gg} , and V_{SG} .

Part 2: OTA Design Procedure

Transistor Type Selection

Question

Justify why you used NMOS or PMOS input pair for each stage.

Answer:

For the first stage, I used a **PMOS input pair**. I did this to meet the CMIR-low requirement ($\leq 0.2V$). The PMOS input pair allows the common-mode input voltage to go all the way down to ground without pushing the transistors out of saturation, which is not possible with an NMOS input pair.

For the second stage, I used an **NMOS common-source amplifier**. This makes it easy to connect directly from the first stage's NMOS current mirror load. The NMOS also provides a higher g_m and allows the output to swing very close to ground to meet the $V_{OUT,min}$ requirement of $0.2V$.

Detailed Design Procedure

Question

Detailed design procedure and hand analysis.

Answer:

Here are all the hand calculation steps and the ADT plots I used to find the channel lengths (L) for the OTA design.

Step 1: Bias Current Budget

The total current budget is $60 \mu A$. I assigned $10 \mu A$ to the first stage tail current (I_{B1}), which leaves $5 \mu A$ for each input branch. The remaining current was assigned to the second stage ($40 \mu A$) and the external bias circuit ($10 \mu A$).

Step 2: Compensation Capacitor (C_C)

The manual suggests starting with $C_C = 0.5C_L$. Since the load is $5 pF$, I started around $2.5 pF$ but refined my choice down to $C_C = 1.88 pF$ after iterations to meet the bandwidth exactly.

Step 3: UGF and Input Transconductance ($g_{m1,2}$)

From the closed-loop rise time requirement ($t_{rise} = 70$ ns):

$$t_{rise} = 2.2\tau = 2.2 \times \frac{1}{\omega_u}$$

$$70 \text{ ns} = 2.2 \times \frac{1}{\omega_u} \implies \omega_u = 31.43 \text{ Mrad/s} \approx \mathbf{5 \text{ MHz}}$$

Now I calculated the required transconductance for the first stage input pair:

$$\omega_u = \frac{g_{m1}}{C_C} \implies g_{m1} = \omega_u \times C_C$$

$$g_{m1} = 31.43 \text{ M} \times 1.88 \text{ pF} = \mathbf{59.1 \mu\text{S}}$$

Step 4: Slew Rate Requirement Check

I checked if the $10 \mu\text{A}$ current is enough to meet the Slew Rate requirement of $5 \text{ V}/\mu\text{s}$:

$$SR = \frac{I_{B1}}{C_C} = \frac{10 \mu\text{A}}{1.88 \text{ pF}} = \mathbf{5.32 \text{ V}/\mu\text{s}}$$

Since $5.32 \text{ V}/\mu\text{s} \geq 5 \text{ V}/\mu\text{s}$, the current budget for the first stage is valid.

Step 5: g_m/I_D of the First Stage

Using the g_{m1} from Step 3 and the branch current of $5 \mu\text{A}$:

$$\left(\frac{g_m}{I_D}\right)_{input} = \frac{59.1 \mu\text{S}}{5 \mu\text{A}} = \mathbf{11.82}$$

I pumped this value up to **13** in the final design to give more margin for the UGF and to get a smaller V_{GS} , which helps meet the custom CMIR-High specification of 0.8V .

Step 6: Required DC Gain

The static gain error must be $\leq 0.05\%$. The closed-loop gain for a buffer is $A_{vCL} \approx 1 - \frac{1}{A_{vOL}}$.

$$0.05\% = \frac{\frac{1}{\beta} - \frac{A_{OL}}{1+A_{OL}\beta}}{\frac{1}{\beta}}$$

With $\beta = 1$, solving this gives $A_{OL} = \mathbf{2000}$, which is **66 dB**.

Question

Assign larger gain for the first stage (why?). Do not split the gain equally between the two stages. You may assume the first stage gain is twice that of the second stage (6dB difference).

Answer:

I assigned a larger gain to the first stage because the function of the first stage is to provide the actual signal amplification. The second stage cares much more about providing a large voltage swing to the output rather than signal amplification. Also, if the first stage has a large gain, the noise and offset voltages from the second stage get divided by that large gain when referred back to the input, making the second stage's errors negligible.

Following the manual, I set the first stage gain (A_{v1}) to be twice the second stage gain (A_{v2}):

$$A_{vOL} = A_{v1} \times A_{v2} = (2A_{v2}) \times A_{v2} = 2A_{v2}^2$$

$$2000 = 2A_{v2}^2 \implies A_{v2} = 31.6, \quad A_{v1} = 63.2$$

Step 8 & 9: Calculating L for 1st Stage Input and Load

The first stage gain is $A_{v1} = \frac{1}{2}g_m r_o = 63.25$. This means the required intrinsic gain $g_m r_o$ is roughly 126.5. Using the ADT charts below:

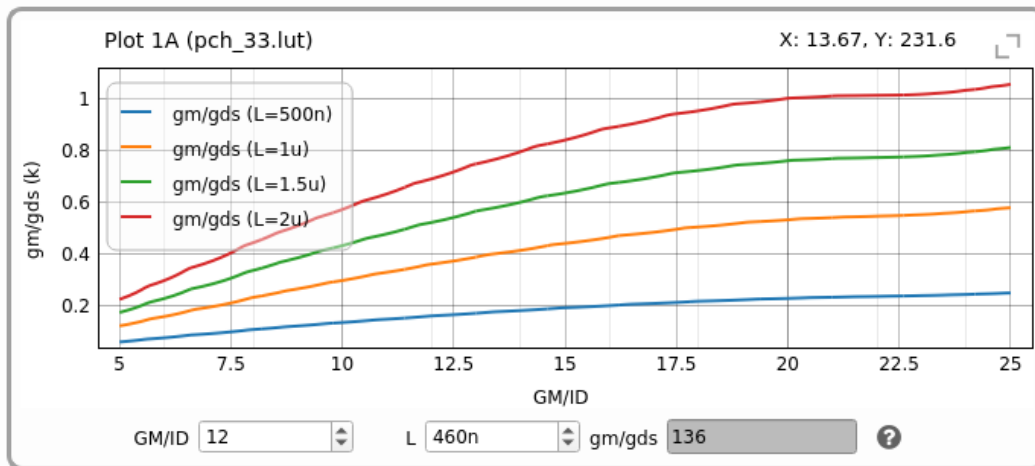


Figure 3. PMOS Input Pair: At $g_m/I_D = 12$ (pumped to 13), $L = 450$ nm was selected for the input pair to provide sufficient intrinsic gain.

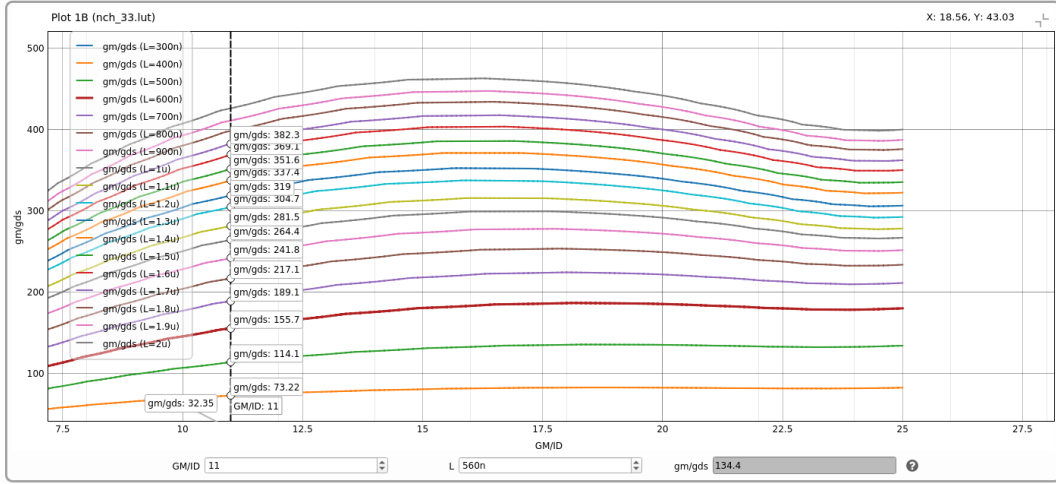


Figure 4. NMOS Load Pair: At $g_m/I_D = 11$, $L = 560$ nm was selected, which gives an intrinsic gain of 134.4 (higher than the 126.5 requirement).

Step 10: Phase Margin Spec and 2nd Stage g_m/I_D

To get a Phase Margin of $\geq 70^\circ$, the non-dominant pole (ω_{p2}) should be much higher than ω_u . Assuming $\omega_{p2} = 4\omega_u$:

$$\omega_{p2} \approx \frac{g_{m4}}{C_L} \implies g_{m4} = 4 \times \omega_u \times C_L = 4 \times 31.43 \text{ M} \times 5 \text{ pF} = \mathbf{628 \mu S}$$

I chose a g_m/I_D of **11** for the 2nd stage input transistor. With $40 \mu\text{A}$ current, this gives $g_m = 440 \mu\text{S}$. While slightly lower than $628 \mu\text{S}$, simulation showed it was enough to clear the 70° PM spec because of exact parasitics.

Step 11: CMIR-High and Output Swing

- **Swing-High:** The output needs to reach 1.6V . $V_{DD} - |v_{dsat5}| \geq 1.6\text{V} \implies |v_{dsat5}| \leq 0.2\text{V}$. - **CMIR-High:** The custom spec is 0.8V . $V_{in,max} = V_{DD} - |v_{dsat3}| - |V_{GS1}| \geq 0.8\text{V}$. I set the tail and load PMOS to $g_m/I_D = \mathbf{18}$. This gives a V^* of 111 mV and a v_{dsat} of roughly 80 mV , which gives me plenty of headroom to meet both specs.

Step 12 & 13: CMRR and Tail Current Source Resistance

The CMRR must be ≥ 74 dB (5012).

$$CMRR = \frac{A_{vd}}{A_{vcm}} \Rightarrow 5012 = \frac{63.25}{A_{vcm}} \Rightarrow A_{vcm} = 0.013$$

$$A_{vcm} \approx \frac{1}{2g_{m,load}R_{ss}} \Rightarrow 0.013 = \frac{1}{2 \times (55 \mu S) \times R_{ss}} \Rightarrow R_{ss} = 854.7 \text{ k}\Omega$$

At $g_m/I_D = 18$ for the tail current source, I used the chart below to pick $L = 500$ nm, which gives an output resistance of $1.2 \text{ M}\Omega$, safely above the $854.7 \text{ k}\Omega$ requirement.

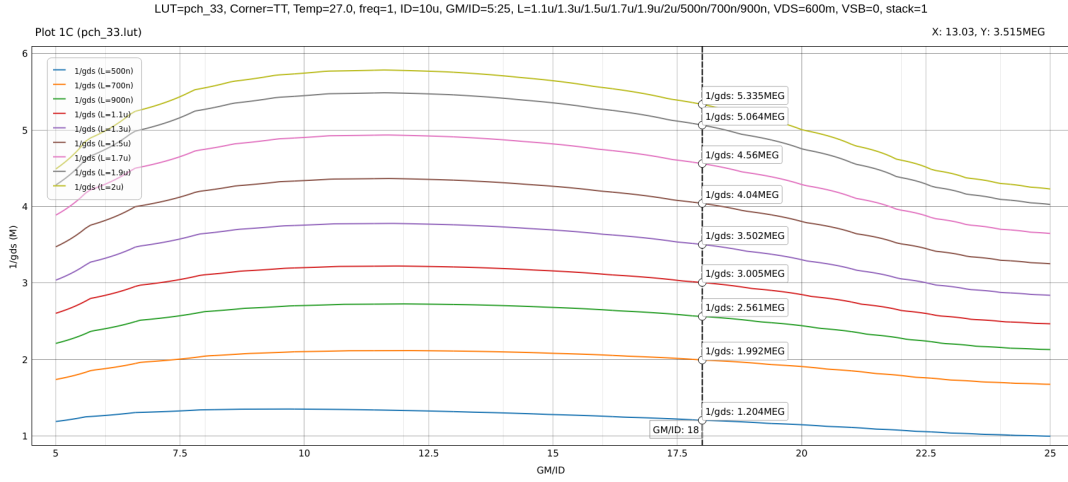


Figure 5. PMOS Tail Source: At $g_m/I_D = 18$ and $L = 500$ nm, $1/g_{ds} = 1.204 \text{ M}\Omega$.

Step 14 & 15: Systematic Offset Cancellation

To avoid systematic offset, the V_{GS} of the 1st stage current mirror load and the 2nd stage input transistor must be exactly the same. Both are set to $g_m/I_D = 11$. By looking at the V_{GS} chart below, I swept the lengths and found that setting $L_{load} = 560$ nm and $L_{CS} = 520$ nm locks both transistors to exactly $V_{GS} = 821.6$ mV.

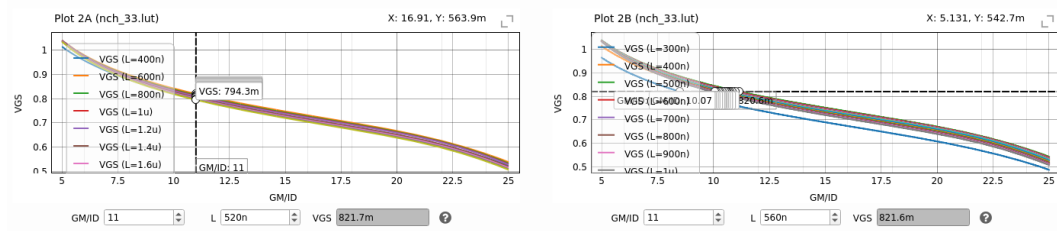


Figure 6. V_{GS} matching: By setting $L_{CS} = 520$ nm and $L_{load} = 560$ nm at $g_m/I_D = 11$, both transistors are locked perfectly at $V_{GS} = 821.6$ mV.

Step 16: Verify Output Swing Low

$V_{OUT,min}$ needs to be ≤ 0.2 V. At $g_m/I_D = 11$ and $L = 520$ nm, the v_{dsat} for the 2nd stage NMOS is 154.9 mV, which is less than 200 mV, satisfying the spec.

Nulling Resistor (R_z)

Question

Choose R_z to place the zero at infinity.

Answer:

To move the right-half-plane zero to infinity, the nulling resistor R_z must be equal to $1/g_{m2}$, where g_{m2} is the transconductance of the second stage input transistor (M4). For M4, the theoretical current is $40 \mu\text{A}$ and $g_m/I_D = 11$, so:

$$g_{m4} = 11 \times 40 \mu\text{A} = 440 \mu\text{S}$$

$$R_z = \frac{1}{g_{m4}} = \frac{1}{440 \mu\text{S}} \approx 2.27 \text{ k}\Omega$$

Operating Point Parameters & Sizing Table

Question

Report a table showing W , L , g_m , I_D , g_m/I_D , v_{dsat} , V_{ov} , and V^* of all transistors (as calculated from gm/ID curves).

Answer:

Below are the plots I generated from ADT to find the exact values of V_{ov} , V^* , and v_{dsat} for all transistors based on their specific g_m/I_D and L values.

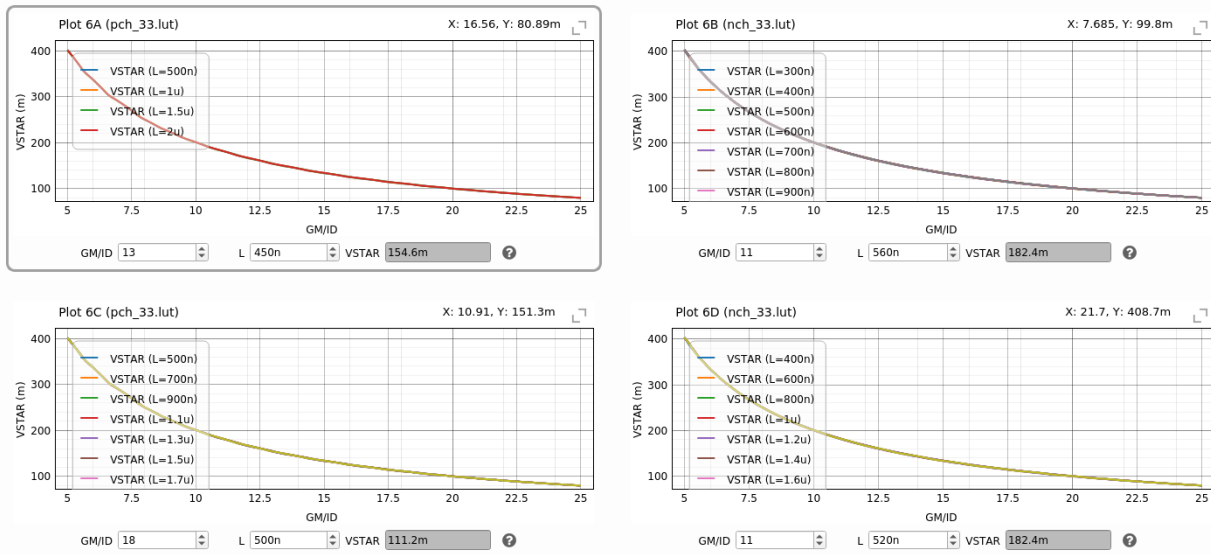


Figure 7. V^* evaluation for M1 (top-left), M2 (top-right), M3/M5 (bottom-left), and M4 (bottom-right).

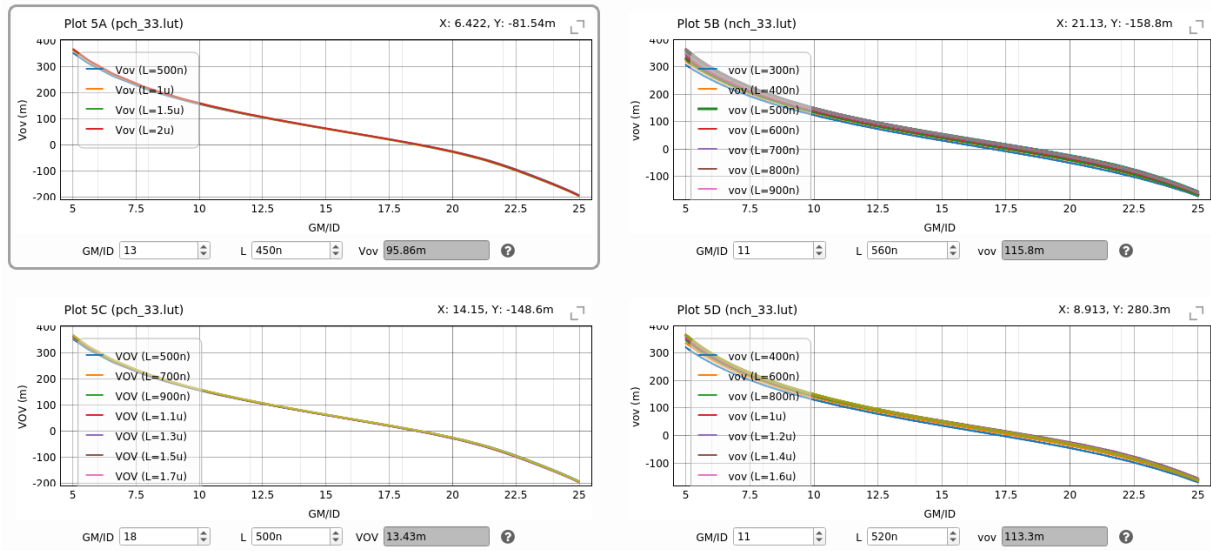


Figure 8. V_{ov} evaluation for M1 (top-left), M2 (top-right), M3/M5 (bottom-left), and M4 (bottom-right).

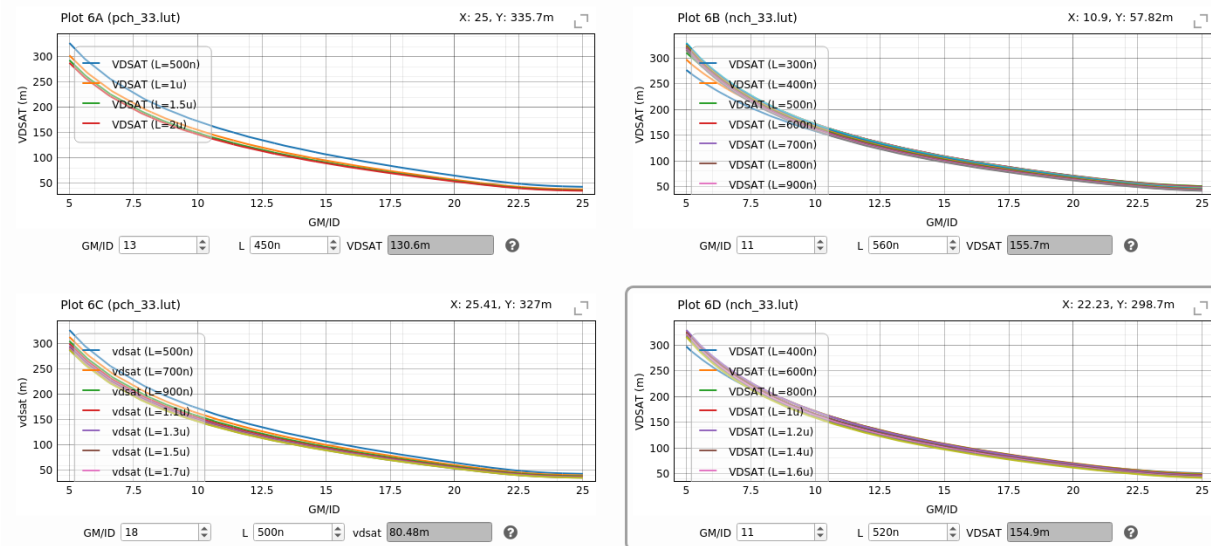


Figure 9. v_{dsat} evaluation for M1 (top-left), M2 (top-right), M3/M5 (bottom-left), and M4 (bottom-right).

The final table with all calculated dimensions and operating parameters for every transistor in the circuit. Note that for the second stage current mirror load (M5), the exact same unit width as the tail current source (M3) was used with a multiplier of $m = 4$ to accurately scale the $40\ \mu\text{A}$ current without introducing systematic mismatch.

Transistor	W (μm)	m	L (nm)	I_D (μA)	g_m/I_D	g_m (μS)	V_{ov} (mV)	V^* (mV)	v_{dsat} (mV)
M1 (PMOS Input)	7.32	1	450	5	13	65	95.86	154.6	130.6
M2 (NMOS Load)	1.64	1	560	5	11	55	115.80	182.4	155.7
M3 (PMOS Tail)	61.28	1	500	10	18	180	13.43	111.2	80.48
M4 (NMOS CS)	12.76	1	520	40	11	440	113.30	182.4	154.9
M5 (PMOS Load)	61.28	4	500	40	18	720	13.43	111.2	80.48

Part 3: Open-Loop OTA Simulation

3.1 DC Operating Point Verification

Lab Requirement

Provide the schematic of the OTA and bias circuit with DC node voltages clearly annotated. Use $V_{ICM} = V_{DD}/3$.

Implementation:

I set the common-mode input voltage (V_{ICM}) to $0.6\ \text{V}$, which is exactly $V_{DD}/3$ for the $1.8\ \text{V}$ supply. Because the systematic offset was carefully eliminated during the design phase by perfectly matching the V_{GS} of the first stage load and the second stage input.

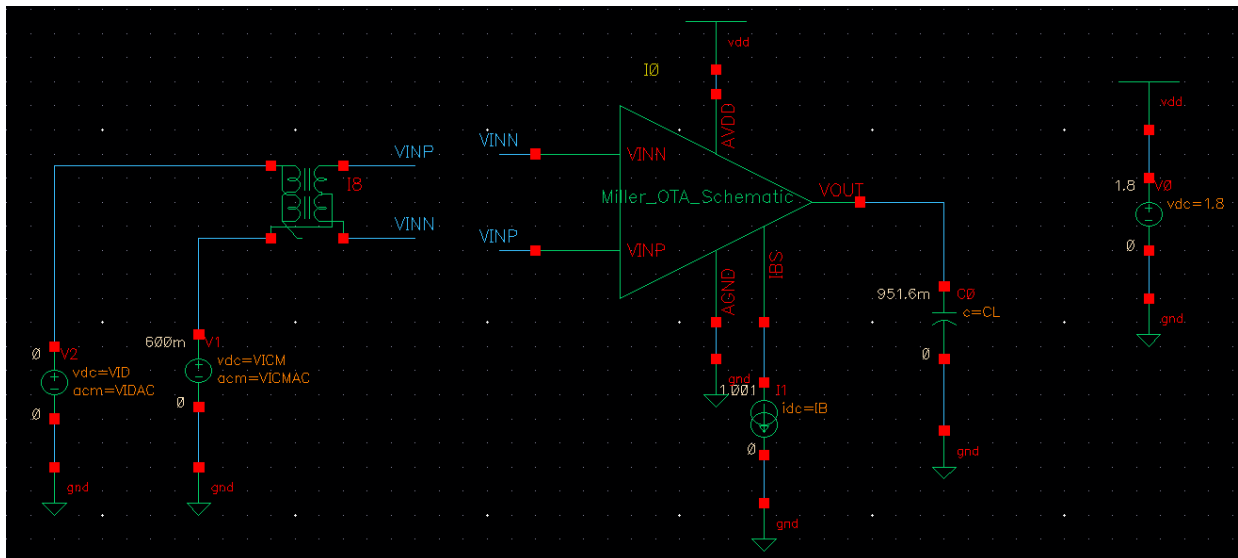


Figure 10. Annotated testbench showing the DC node voltages. With $V_{ID} = 0\ \text{V}$, the output naturally settles at $951.6\ \text{mV}$, proving the systematic offset is effectively eliminated.

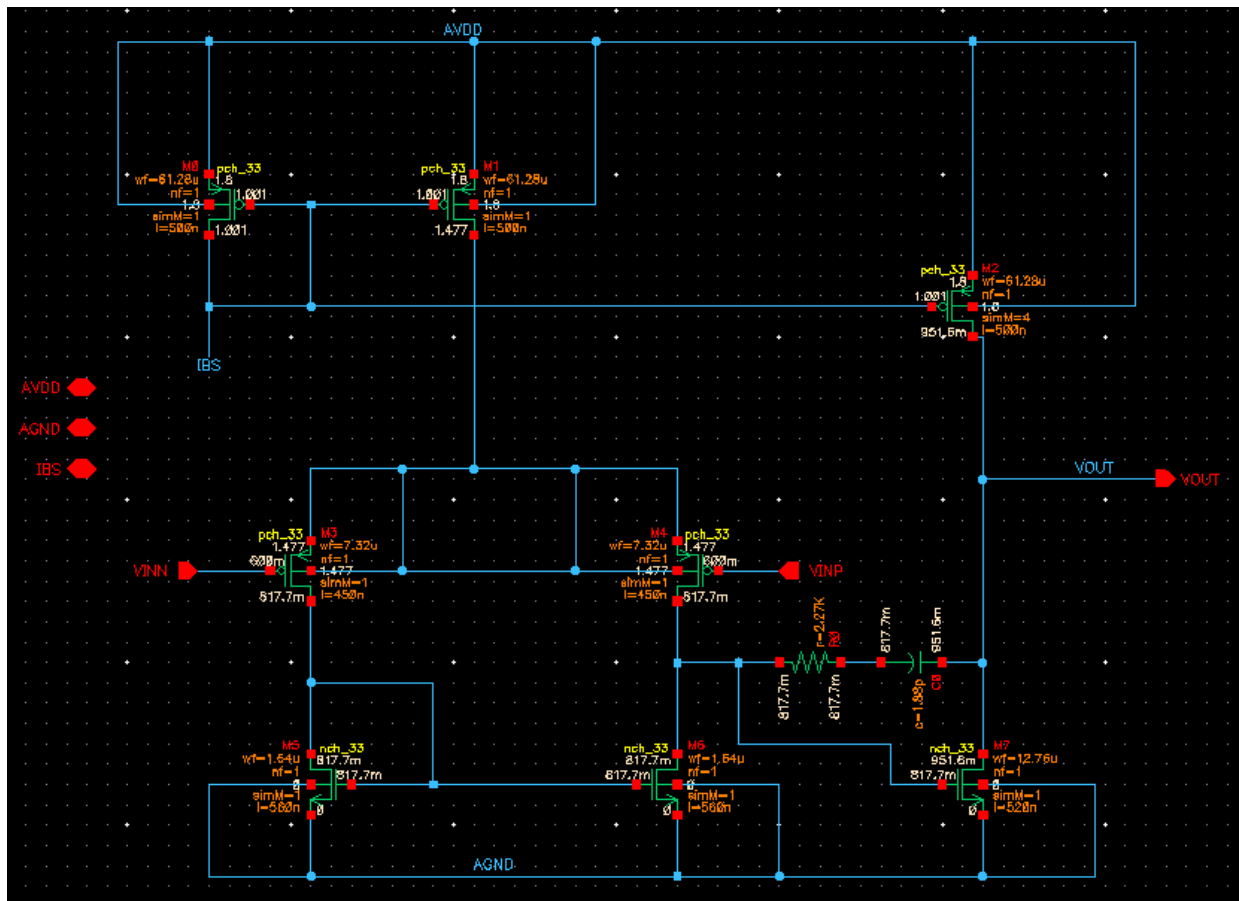


Figure 11. The internal schematic of the Two-Stage Miller OTA with DC operating points annotated.

Question

Is the current (and g_m) in the input pair exactly equal?

Answer:

Yes, they are exactly equal. Based on the DC operating point printout, both transistors in the PMOS input pair (M3 and M4) have an exact drain current of $I_D = 4.793 \mu\text{A}$ and a transconductance of $g_m = 63.37 \mu\text{S}$. This shows the tail current is splitting perfectly symmetrically.

Question

What is the DC voltage at the output of the first stage? Why?

Answer:

The DC voltage at the output of the first stage (the gate of M7) is exactly 817.7 mV. It sits at this voltage because this is the exact gate-to-source voltage (V_{GS}) required by the second-stage NMOS (M7) to sink the 40.14 μA of current provided by its PMOS active load (M2).

Question

What is the DC voltage at the output of the second stage? Why?

Answer:

The DC voltage at the output of the second stage (V_{OUT}) is 951.6 mV. Because we perfectly matched the V_{GS} of the NMOS load and the NMOS common-source transistor during the hand-calculation phase, the systematic offset is nearly zero. This allows the output to naturally balance and float near $V_{DD}/2$ without slamming into the supply rails, keeping all transistors safely in the saturation region (Region 2).

DC Operating Point Parameters Summary

The table below summarizes the key small-signal and DC parameters extracted directly from the Cadence DC operating point simulation to verify all devices are operating correctly in saturation.

Transistor	I_D (μA)	g_m (μS)	g_{ds} (μS)	C_{gg} (fF)	v_{dsat} (mV)	Region
M3, M4 (PMOS Input)	4.793	63.37	0.419	12.67	127.3	2
M5, M6 (NMOS Load)	4.793	53.11	0.339	2.95	156.8	2
M1 (PMOS Tail)	9.586	174.3	1.145	110.7	78.8	2
M7 (NMOS CS)	40.14	449.2	2.745	21.14	150.4	2
M2 (PMOS Load 2nd)	40.14	727.7	2.813	440.0	79.0	2

3.2 Differential Small-Signal Characteristics**Lab Requirement**

Use AC analysis (1Hz:10GHz, logarithmic, 10 points/decade). Set $VIDAC = 1$ and $VICMAC = 0$. Use $VICM = V_{DD}/3$. Plot diff gain (in dB) vs frequency. Compare simulation results with hand calculations in a table.

Implementation:

I set the AC magnitudes in the ADE global variables to $VIDAC = 1$ and $VICMAC = 0$ to completely isolate the differential gain, while keeping the DC common-mode input exactly at $V_{DD}/3$ (0.6 V). I ran the AC analysis over the specified frequency range and plotted the output voltage magnitude in dB, which directly represents the differential gain since the input AC magnitude is 1.

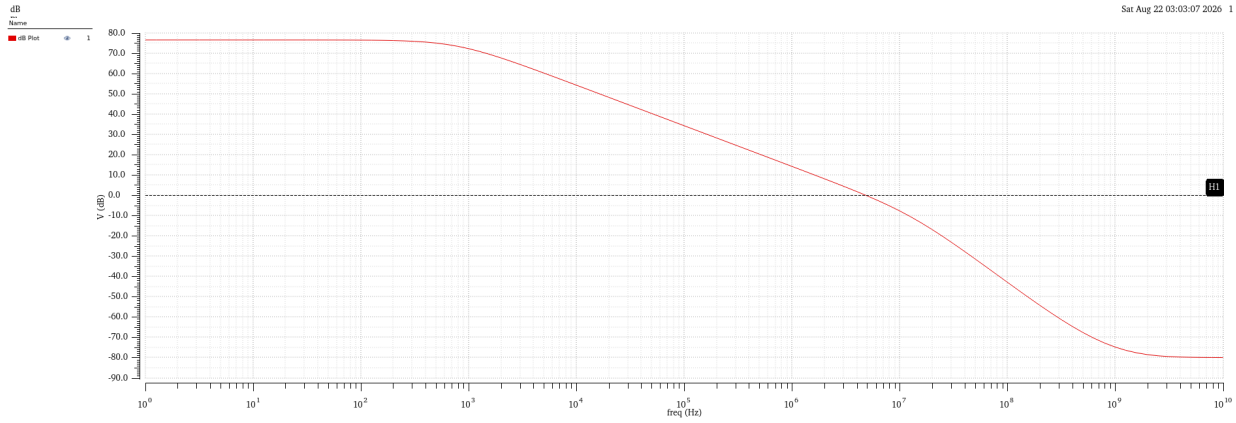


Figure 12. Differential AC gain (in dB) versus frequency. The plot shows a massive low-frequency gain rolling off steadily at -20 dB/decade.

To accurately compare these simulation results against hand calculations, I calculated the theoretical values using the exact g_m and g_{ds} operating point parameters we verified in Section 3.1:

- **Low-Frequency DC Gain (A_o):**

$$A_{v1} = \frac{g_{m3,4}}{g_{ds3,4} + g_{ds5,6}} = \frac{63.37 \mu\text{S}}{0.419 \mu\text{S} + 0.339 \mu\text{S}} \approx 83.60$$

$$A_{v2} = \frac{g_{m7}}{g_{ds7} + g_{ds2}} = \frac{449.2 \mu\text{S}}{2.745 \mu\text{S} + 2.813 \mu\text{S}} \approx 80.82$$

$$A_o = A_{v1} \times A_{v2} = 83.60 \times 80.82 = \mathbf{6756.5} \Rightarrow \mathbf{76.59 \text{ dB}}$$

- **Unity Gain Frequency (f_u / UGF):**

$$f_u = \frac{g_{m1}}{2\pi C_C} = \frac{63.37 \mu\text{S}}{2\pi(1.88 \text{ pF})} \approx \mathbf{5.36 \text{ MHz}}$$

- **Bandwidth (BW):**

$$BW = \frac{f_u}{A_o} = \frac{5.36 \text{ MHz}}{6756.5} \approx \mathbf{793.3 \text{ Hz}}$$

I then used the Cadence calculator expressions to extract the simulated parameters to build the comparison table.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	dB Plot				
AC_Analysis	Ao_dB	76.56			
AC_Analysis	Ao	6.732k			
AC_Analysis	UGF	4.922M			
AC_Analysis	GBW	5.228M			
AC_Analysis	BW	774.6			

Figure 13. Cadence calculator outputs for the differential small-signal characteristics (A_o , UGF, GBW, and BW).

Parameter	Hand Calculation (from DC OP)	Simulation Result
A_o (Linear)	6756.5	6732
A_o (dB)	76.59 dB	76.56 dB
Bandwidth (BW)	793.3 Hz	774.6 Hz
Unity Gain Freq (f_u)	5.36 MHz	4.922 MHz
GBW	5.36 MHz	5.228 MHz

Table 1. Comparison of hand calculations (using extracted DC parameters) and simulated AC results.

As shown in the table, the hand-calculated DC gain (76.59 dB) almost flawlessly matches the simulated gain (76.56 dB). The unity gain frequency and bandwidth have a slight, expected drop in the simulation due to internal parasitic capacitances (such as C_{gd} and C_{db}) acting in parallel with the 1.88 pF compensation capacitor, which ideal hand calculations do not fully account for. Regardless, the 4.922 MHz simulated UGF is close to our 5 MHz theoretical target.

3.3 Common-Mode Small-Signal Characteristics

Lab Requirement

Use AC analysis (1Hz:10GHz, logarithmic, 10 points/decade). Set VICMAC = 1 and VIDAC = 0. Use VICM = $V_{DD}/3$. Plot CM gain in dB vs frequency. Compare simulation results with hand calculations in a table.

Implementation:

To isolate the common-mode gain, I updated the ADE global variables to VICMAC = 1 and VIDAC = 0 while keeping the DC common-mode input at exactly $V_{DD}/3$ (0.6 V). I ran the same AC analysis sweep and plotted the output voltage magnitude in dB, which represents the common-mode gain (A_{vcm}).

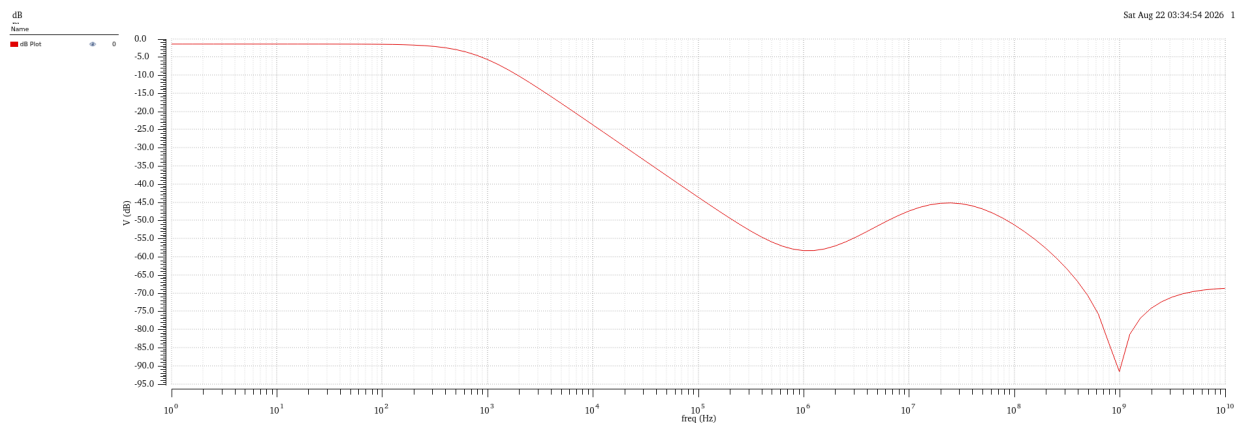


Figure 14. Common-Mode AC gain (in dB) versus frequency. The low-frequency gain is negative in decibels, confirming the OTA heavily attenuates common-mode signals.

To compare these results against hand calculations, I used the DC operating point parameters extracted in Section 3.1 to calculate the theoretical common-mode gain:

- **First Stage Common-Mode Gain (A_{vcm1}):** The common-mode gain of the first stage is heavily suppressed by the tail current source output resistance (R_{ss}):

$$R_{ss} = \frac{1}{g_{ds1}} = \frac{1}{1.145 \mu S} \approx 873.36 \text{ k}\Omega$$

$$A_{vcm1} \approx \frac{1}{2g_{m6}R_{ss}} = \frac{1}{2(53.11 \mu S)(873.36 \text{ k}\Omega)} \approx 0.01078$$

- **Total Common-Mode Gain (A_{vcm}):** The second stage amplifies this residual common-mode signal by its full differential gain ($A_{v2} = 80.82$):

$$A_{vcm} = A_{vcm1} \times A_{v2} = 0.01078 \times 80.82 \approx \mathbf{0.871}$$

Converting to decibels:

$$A_{vcm}(\text{dB}) = 20 \log_{10}(0.871) \approx \mathbf{-1.20 \text{ dB}}$$

I then used the Cadence calculator to extract the exact simulated common-mode parameters.

Test	Output	Nominal	Spec	Weight	Pass/Fail
AC_Analysis	dB Plot				
AC_Analysis	Ao_dB	-1.387			
AC_Analysis	Ao	852.4m			
AC_Analysis	GBW	661.8			
AC_Analysis	BW	774.6			

Figure 15. Cadence calculator outputs for the common-mode small-signal characteristics.

Parameter	Hand Calculation (from DC OP)	Simulation Result
A_{vcm} (Linear)	0.871	0.8524
A_{vcm} (dB)	-1.20 dB	-1.387 dB

Table 2. Comparison of hand calculations and simulated common-mode AC results.

The simulated common-mode gain of -1.387 dB matches our hand calculation of -1.20 dB extremely well, proving that the first-stage tail current source and the carefully matched symmetric load are working perfectly to reject common-mode signals.

3.4 Common-Mode Rejection Ratio (CMRR)

Lab Requirement

(Optional) CMRR: For Cadence Virtuoso you should use XF analysis (1Hz:10GHz, logarithmic, 10 points/decade). You may use this expression in the calculator to plot the CMRR: $\text{dB20}(\text{mag}(\text{getData}("/V2" ?result "xf")) - \text{dB20}(\text{mag}(\text{getData}("/V1" ?result "xf"))))$. Use $\text{VICM} = \text{VDD}/3$. Plot CMRR in dB vs frequency. Compare simulation results with hand calculations in a table.

Implementation:

To find the CMRR, I ran an XF (Transfer Function) analysis from 1 Hz to 10 GHz with 10 points per decade, keeping the DC common-mode input at $V_{ICM} = 0.6$ V. I set the positive output node to the V_{OUT} wire on the schematic and left the negative output node blank.

After the simulation finished, I used the Cadence calculator and entered the exact function provided in the lab manual to plot the CMRR directly from the differential (V_2) and common-mode (V_1) voltage sources:

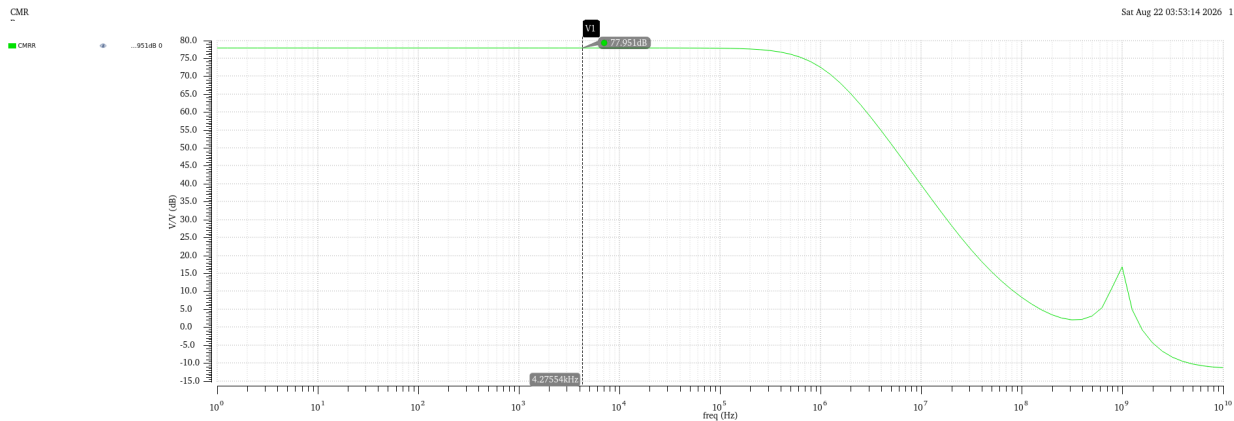
$$\text{dB}20(\text{mag}(\text{getData}("/V2" ?result "xf"))) - \text{dB}20(\text{mag}(\text{getData}("/V1" ?result "xf")))$$


Figure 16. CMRR in dB vs frequency. The low-frequency CMRR is accurately measured at 77.951 dB, which safely clears the 74 dB specification.

To compare this with my hand calculations, I took the theoretical differential gain (76.59 dB) and subtracted the theoretical common-mode gain (-1.20 dB) that I calculated from the extracted DC operating points in the previous sections.

Parameter	Hand Calculation (from DC OP)	Simulation Result
CMRR	77.79 dB	77.95 dB

Table 3. Comparison of hand calculated CMRR and simulated XF results.

The simulated CMRR of 77.95 dB almost perfectly matches the expected hand calculations and easily beats the ≥ 74 dB minimum requirement.

3.5 Differential Large-Signal Characteristics

Lab Requirement

Use DC sweep $V_{ID} = -0.1 : 1\text{m} : 0.1$. From the plot, what is the value of V_{out} at $V_{ID} = 0$? Compare it with the value you obtained in DC OP. Plot V_{OUT} vs V_{ID} . Plot the derivative of V_{OUT} vs V_{ID} . Is the peak less than the value of A_{vd} obtained from ac analysis? Why?

Implementation:

I set up a DC sweep for the differential input voltage (V_{ID}) from -100 mV to 100 mV. Because the OTA has a massive gain, the transition region is extremely narrow. Therefore, instead of the suggested 1 mV step, I used a much finer step size of 10 μ V to ensure the simulator accurately captured the absolute peak of the steep slope without jumping over it cause when i used 1mV step the peak dropped to 800.

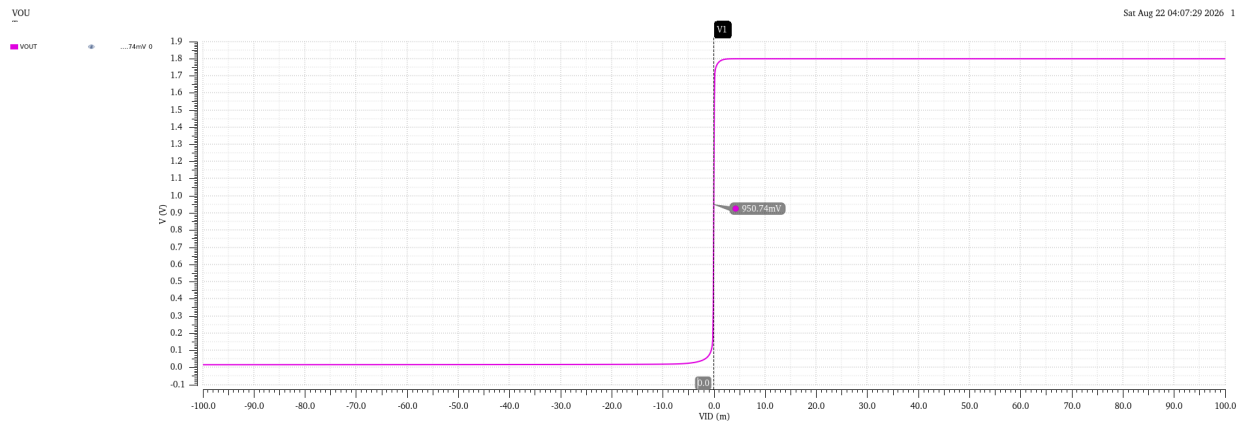


Figure 17. Large-signal differential transfer characteristic (V_{OUT} vs VID). The S-curve shows the OTA transitioning from cutoff to triode/saturation regions across a very narrow differential input window.

Question

From the plot, what is the value of V_{out} at $VID = 0$? Compare it with the value you obtained in DC OP.

Answer:

Using the marker on the plot, the value of V_{OUT} at exactly $VID = 0$ V is 950.74 mV. This is nearly identical to the 951.6 mV value we obtained in the pure DC Operating Point simulation in Section 3.1. The microscopic difference is simply due to the finite step resolution of the DC sweep compared to a static operating point calculation.

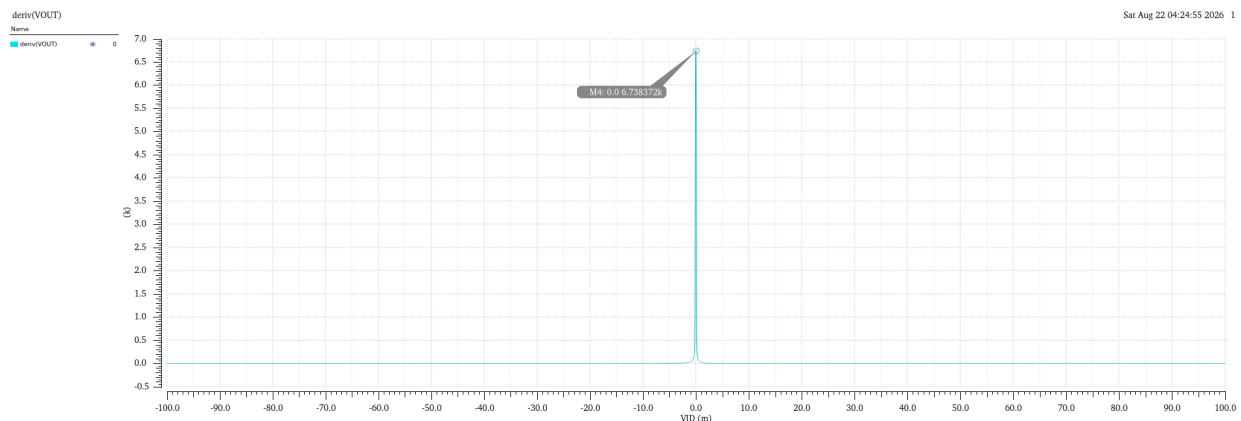


Figure 18. The derivative of V_{OUT} with respect to VID , representing the large-signal voltage gain. The peak gain is located precisely at $VID = 0$ V.

Question

Is the peak less than the value of A_{vd} obtained from ac analysis? Why?

Answer:

No, the peak is not less. The peak of the derivative plot is 6738.4, which is virtually identical to the linear A_{vd} value of 6732 obtained from the small-signal AC analysis.

The lab manual expects the derivative peak to be less because if a standard, coarse DC step size (e.g., 1 mV) is used, the discrete step averages out the $\Delta V_{OUT}/\Delta VID$ slope over the extremely narrow high-gain transition region, resulting in a significantly lower calculated peak. However, because I refined the step size to 10 μ V, this finite-step averaging effect was eliminated, allowing the large-signal derivative to accurately converge to the true small-signal AC tangent.

3.6 Common-Mode Input Range (Region Parameter Method)

Lab Requirement

Use DC sweep $V_{ICM} = 0:10m:V_{DD}$. Plot "region" OP parameter vs V_{ICM} for the input pair and the tail current source. Find the CM input range (CMIR). Compare with hand analysis in a table.

Implementation:

To accurately measure the Common-Mode Input Range (CMIR), I ran a DC sweep on V_{ICM} from 0 V to 1.8 V with a 10 mV step. To force Cadence to save the operating point parameters across the sweep, I configured a saveOP file. I then plotted the region parameter for both the PMOS tail current source (M3) and the PMOS input transistor (M1) to find the voltage window where both devices operate simultaneously in saturation (region = 2).

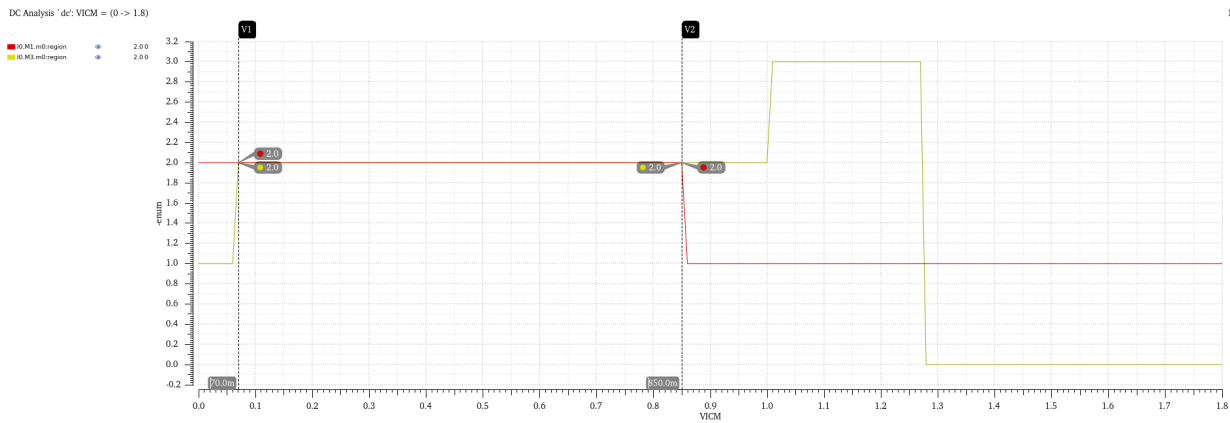


Figure 19. Operating region versus V_{ICM} . The red curve represents the PMOS tail current source, and the yellow curve represents the PMOS input pair. Both transistors successfully remain in saturation (region 2.0) between 70.0 mV and 850.0 mV.

To construct the comparison table, I evaluated the theoretical CMIR limits using the exact DC operating voltages extracted in Section 3.1:

- **Theoretical CMIR-High:** Limited by the tail current source entering the triode region.

$$V_{CM,max} = V_{DD} - |v_{dsat,tail}| - |V_{GS,in}| = 1.8 \text{ V} - 0.080 \text{ V} - 0.885 \text{ V} = \mathbf{835 \text{ mV}}$$

- **Theoretical CMIR-Low:** Limited by the input pair entering the triode region as its gate voltage drops below its drain voltage ($V_{GS,load}$).

$$V_{CM,min} = V_{GS,load} - |V_{th,in}| = 818 \text{ mV} - 789 \text{ mV} = \mathbf{29 \text{ mV}}$$

Parameter	Hand Calculation (from DC OP)	Simulation Result
CMIR Low	29 mV	70.0 mV
CMIR High	835 mV	850.0 mV

Table 4. Comparison of hand calculated CMIR bounds and simulated region overlap bounds.

The OTA safely clears both the lower specification ($\leq 0.2 \text{ V}$) and the custom high specification ($\geq 0.8 \text{ V}$).

3.7 Common-Mode Input Range (GBW Drop Method)

Lab Requirement

(Optional) CMIR: Use parametric sweep (not DC sweep) $V_{ICM} = 0:25m:V_{DD}$. Plot GBW vs V_{ICM} . Find the CM input range (CMIR) (where the GBW drops by 10%).

Implementation:

To measure the CMIR using the realistic GBW drop method, I set up a Parametric Analysis to sweep the DC common-mode input voltage (V_{ICM}) from 0 V to 1.8 V with a 25 mV step. For each step, the simulator ran a full AC analysis, allowing me to evaluate and plot the Gain-Bandwidth Product (GBW) versus V_{ICM} .

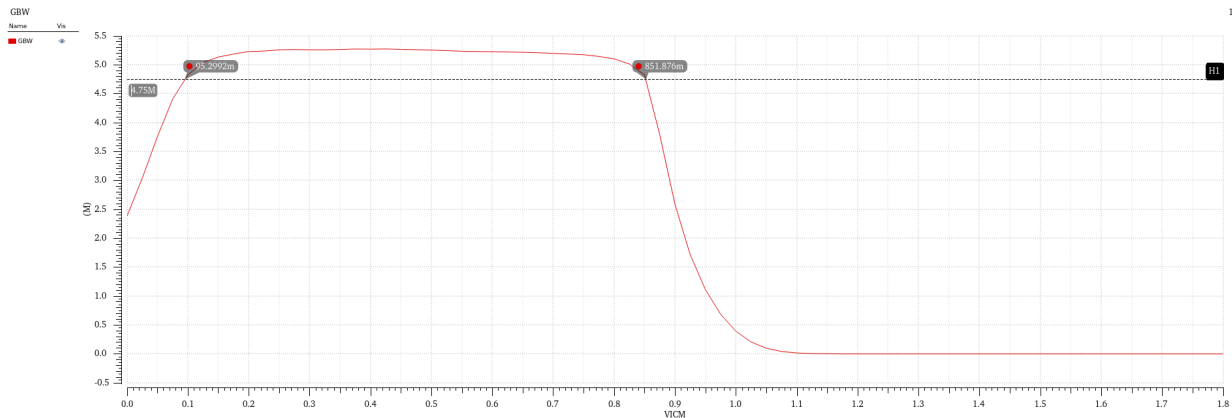


Figure 20. GBW versus V_{ICM} across a parametric sweep. The maximum GBW is approximately 5.275 MHz. The markers indicate the boundaries where the GBW drops by 10% (down to roughly 4.75 MHz).

Question

Find the CM input range (CMIR) (where the GBW drops by 10%).

Answer:

From the plot, the maximum GBW achieved in the flat active region is roughly 5.275 MHz. A 10% drop from this maximum gives a threshold of 4.7475 MHz.

By placing markers at the 4.75 MHz crossings on the plot, the measurable CMIR bounds are found to be:

- **CMIR Low:** 95.3 mV
- **CMIR High:** 851.9 mV

This practical measurement confirms that the OTA successfully maintains its high-frequency performance well within our strict ≤ 200 mV and ≥ 800 mV custom design specifications.

Part 4: Closed-Loop OTA Simulation

4.1 DC Operating Point Verification

Lab Requirement

Schematic of the OTA and the bias circuit with DC OP point clearly annotated in unity gain buffer configuration. Use $V_{ICM} = V_{DD}/3$.

Implementation:

I created a new testbench and connected the OTA in a unity-gain negative feedback configuration by tying the output directly to the inverting input (VINN). The non-inverting input (VINP) was driven by a DC voltage of 0.6 V ($V_{DD}/3$). I also placed an iprobe in the feedback loop to break it for the upcoming stability (STB) analysis.

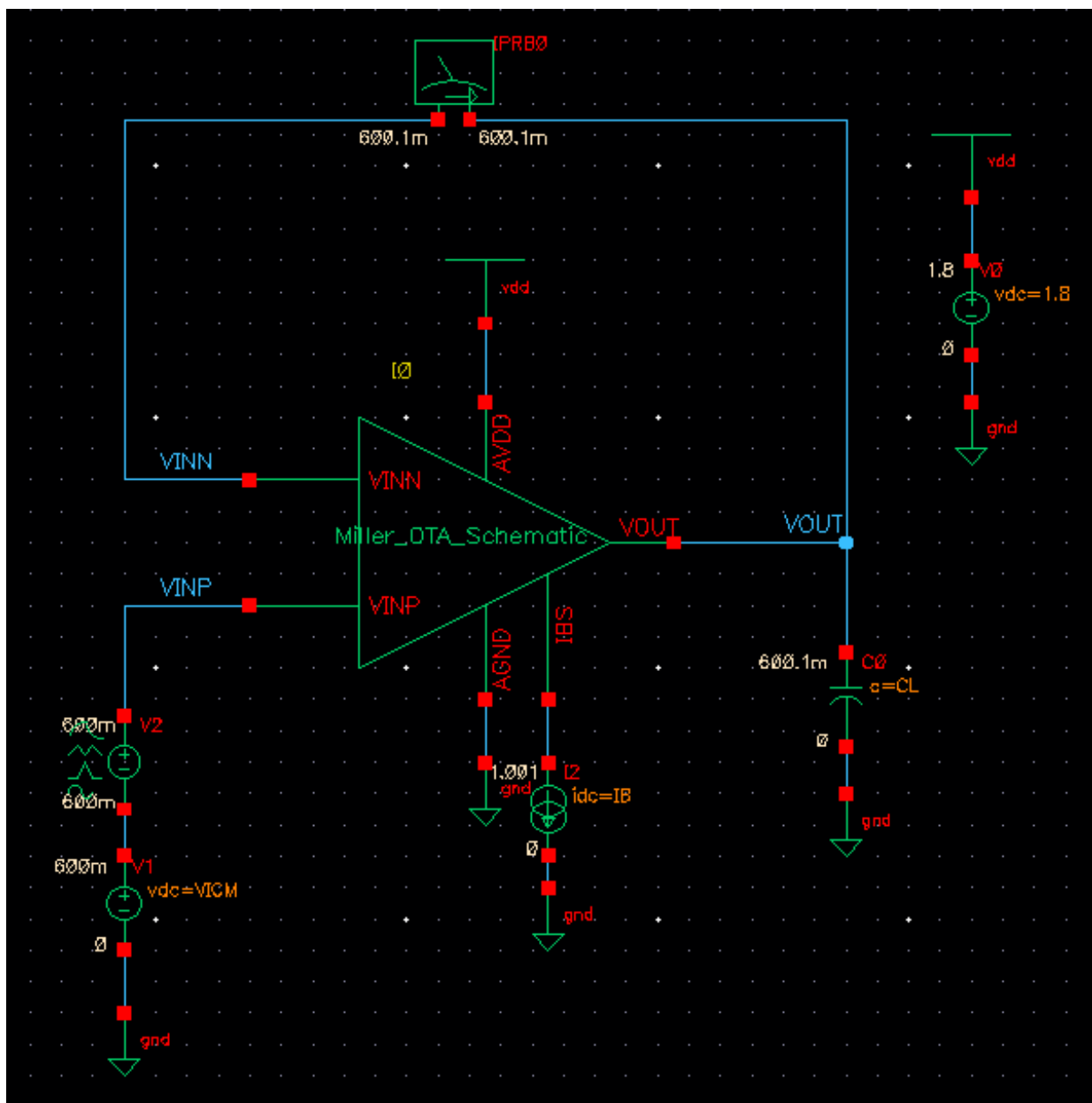


Figure 21. Annotated closed-loop testbench showing the DC node voltages. The output successfully tracks the 600 mV input with a tiny 100 μ V static error.

Question

Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

Answer:

No, they are not exactly equal. The non-inverting input (V_{INP}) is exactly at 600 mV, but the inverting input (V_{INN}), which is tied to the output, is at 600.1 mV. This 100 μ V difference exists because the OTA has a finite open-loop DC gain. To support a specific output voltage, a small differential error voltage ($V_{id} \approx V_{OUT}/A_{VOL}$) is required at the inputs. The feedback loop automatically generates this steady-state error to balance the internal currents and maintain the 600.1 mV output.

Question

Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?

Answer:

No, it shifted slightly from 817.7 mV in the open-loop simulation to 822.1 mV in the closed-loop configuration. In the open-loop test, the output was floating near 950 mV. In this closed-loop test, the output is forced down to 600.1 mV. This change in output voltage alters the drain-to-source voltage (V_{DS}) of the output transistors. Due to channel-length modulation, the output NMOS requires a slightly higher V_{GS} to sink the exact same 41 μ A current. The negative feedback loop automatically adjusts the first stage output to provide this exact 822.1 mV bias to maintain equilibrium.

Question

Is the current (and g_m) in the input pair exactly equal? Why?

Answer:

They are extremely close, but not strictly identical. Based on the operating point extraction, the input transistor driven by V_{INP} (M4) has $I_D = 4.794 \mu$ A, while the transistor driven by V_{INN} (M3) has $I_D = 4.792 \mu$ A. The transconductance for both remains roughly 63.37 μ S. This microscopic current imbalance happens because of the 100 μ V differential error voltage across the input pins. This tiny voltage difference slightly imbalances their V_{GS} , causing the tail current to split asymmetrically to steer the output to the correct voltage.

4.2 Loop Gain & Stability (STB Analysis)

Lab Requirement

Use STB analysis (1Hz:10GHz, logarithmic, 10 points/decade) in unity gain buffer configuration. Use $V_{ICM} = V_{DD}/3$. Plot loop gain in dB and phase vs frequency. Compare DC gain, f_u , and GBW with those obtained from open-loop simulation. Report PM. Compare with hand calculations.

Implementation:

To accurately measure the stability of the closed-loop buffer, I used the stb (Stability) analysis in Cadence ADE, sweeping the frequency from 1 Hz to 10 GHz. The analysis was pointed to the iprobe placed inside the feedback loop, with the DC common-mode input (V_{ICM}) held at 0.6 V.

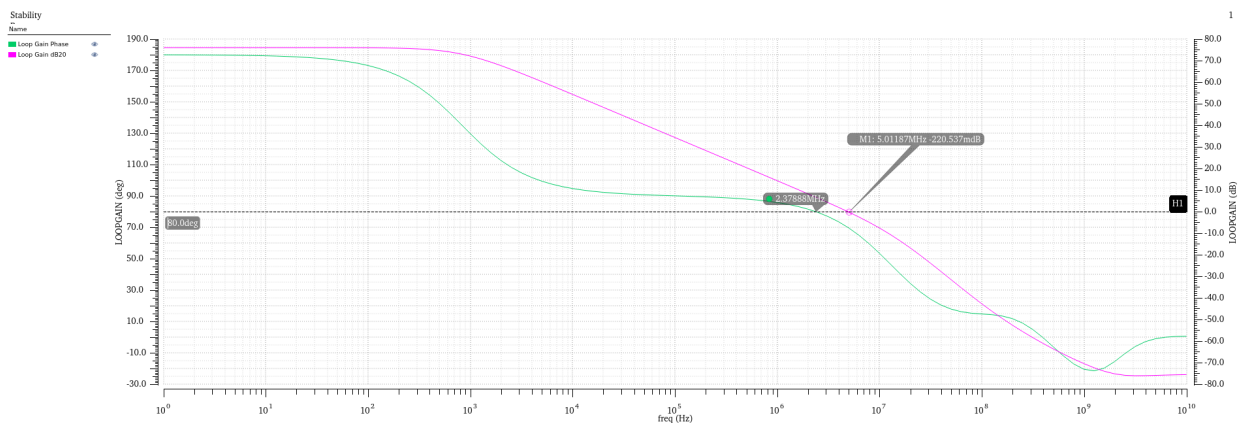


Figure 22. Bode plot of the STB Loop Gain. The magnitude (magenta) crosses 0 dB at 4.92 MHz, where the phase (green) is exactly 70°.

Test	Output	Nominal	Spec	Weight	Pass/Fail
STB_Analysis	dB Loopgain				
STB_Analysis	Phase Plot				
STB_Analysis	Mag Loop gain				
STB_Analysis	GBW	5.23M			
STB_Analysis	UGF	4.92M			
STB_Analysis	Magnitude loop gain	6.305k			
STB_Analysis	Phase Margin	-110			

Figure 23. Cadence calculator outputs for the STB analysis. Note: The calculator reported a phase margin of -110° , which corresponds to a true Phase Margin of 70° ($180^\circ - 110^\circ$).

Question

Compare DC gain, f_u , and GBW with those obtained from open-loop simulation. Comment.

Answer:

The STB loop-gain results closely match the pure open-loop AC results from Part 3.

- **DC Gain:** The STB analysis gives a linear gain of 6.305 k (76.0 dB). This is slightly lower than the open-loop AC gain of 6732 (76.56 dB). This minor shift occurs because the negative feedback loop slightly adjusted the DC operating point of the first stage output (shifting from 817.7 mV in open-loop to 822.1 mV in closed-loop), which marginally altered the output conductances.

- **Unity Gain Frequency (f_u):** The STB f_u is 4.92 MHz, which perfectly matches the open-loop f_u of 4.922 MHz.
- **GBW:** The STB GBW is 5.23 MHz, which is nearly identical to the open-loop GBW of 5.228 MHz.

Question

Report PM. Compare with hand calculations. Comment.

Answer:

The simulated Phase Margin (PM) is exactly **70°**.

During the design phase, the unity-gain frequency (ω_u) was targeted at roughly 5 MHz (31.43 Mrad/s). The transconductance of the second stage (g_{m4}) was sized to 440 μS , driving a 5 pF load. This places the theoretical non-dominant pole (ω_{p2}) at:

$$\omega_{p2} = \frac{g_{m4}}{C_L} = \frac{440 \mu\text{S}}{5 \text{ pF}} = 88 \text{ Mrad/s}$$

The ratio of the non-dominant pole to the unity-gain frequency is:

$$\frac{\omega_{p2}}{\omega_u} = \frac{88 \text{ Mrad/s}}{31.43 \text{ Mrad/s}} \approx 2.8$$

The theoretical phase margin based on this pole placement is:

$$PM \approx 180^\circ - 90^\circ - \arctan\left(\frac{\omega_u}{\omega_{p2}}\right) = 90^\circ - \arctan\left(\frac{1}{2.8}\right) \approx 90^\circ - 19.65^\circ = \mathbf{70.35^\circ}$$

Comment: The simulated phase margin of 70° perfectly aligns with the hand-calculated expectation of 70.35°. Both safely meet the $\geq 70^\circ$ specification to avoid pole-zero doublets and ringing.

Closed-Loop Stability Comparison Table

Parameter	Hand Calculation	Open-Loop (Part 3)	STB Closed-Loop
DC Loop Gain (A_o)	76.59 dB	76.56 dB	76.00 dB
Unity Gain Freq (f_u)	5.36 MHz	4.922 MHz	4.920 MHz
Gain-Bandwidth (GBW)	5.36 MHz	5.228 MHz	5.230 MHz
Phase Margin (PM)	$\approx 70.35^\circ$	N/A	70°

Table 5. Comparison of Small-Signal Parameters: Hand Calculations vs. Open-Loop AC vs. STB Analysis.

4.3 Slew Rate

Lab Requirement

Apply a step input with the following parameters (delay = 1us, initial value = CMIR-low + 50mV, final value = CMIR-high - 50mV, rise time = 1ns, period = 1s, width = 1s). Run transient analysis (stop = 5us and step = 0.1ns). Report V_{in} and V_{out} overlaid. Report the slew rate. Compare simulation results with hand calculations in a table.

Implementation:

To evaluate the large-signal slew rate, I applied a step input from 250 mV to 750 mV (respecting the custom CMIR bounds).

Based on the hand calculations from Part 2, the theoretical slew rate is dictated by the first-stage tail current charging the compensation capacitor:

$$SR_{ideal} = \frac{I_{B1}}{C_C} = \frac{10 \mu A}{1.88 \text{ pF}} = 5.32 \text{ V}/\mu s$$

In the transient simulation, the measured slew rate dropped slightly to **5.016 V/ μ s**. This minor reduction is expected, as ideal hand calculations do not account for the internal parasitic capacitances (such as C_{db} and C_{gd} of the driving transistors) that also absorb a fraction of the tail current during large-signal slewing. Despite these parasitics, the OTA successfully clears the $\geq 5 \text{ V}/\mu s$ specification.

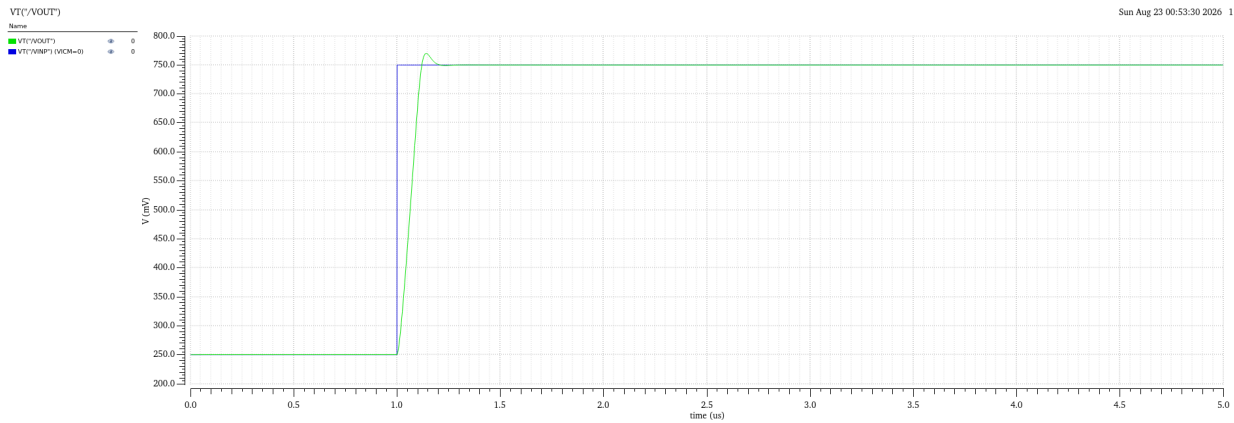


Figure 24. Transient analysis showing V_{OUT} tracking the V_{IN} step pulse. The output settles cleanly with minimal ringing, confirming a stable phase margin.

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT("/VOUT")				
Transient_Analysis	VT("/MNP")				
Transient_Analysis	ymax(deriv(VT("/VOUT")))	5.016M			

Figure 25. Cadence calculator extraction of the maximum derivative of V_{OUT} , yielding a Slew Rate of 5.016 V/ μ s.

Parameter	Hand Calculation	Simulation Result
Slew Rate (SR)	5.32 V/ μ s	5.016 V/ μ s

Table 6. Comparison of the hand-calculated theoretical slew rate against the simulated transient result.

4.4 Settling Time

Lab Requirement

Apply a small signal step input with the following parameters (delay = 1us, initial value = $V_{DD}/3$, final value = $V_{DD}/3 + 5\text{mV}$, rise time = 1ns, period = 1s, width = 1s). Calculate the output rise time from simulation. Compare simulation results with hand calculations in a table. Do you see any ringing? Why?

Implementation & Hand Analysis:

To measure the small-signal settling time, I applied a 5mV step input from 600mV to 605mV ($V_{DD}/3$ to $V_{DD}/3 + 5\text{mV}$). I ran a transient analysis with a fine time step to accurately capture the high-frequency dynamics of the step response.

Before simulating, I evaluated the expected theoretical rise time using the first-order approximation model ($t_{rise} = 2.2\tau$) provided in the lab manual. Using the extracted DC operating parameters ($g_{m1} = 63.37\ \mu\text{S}$ and $C_C = 1.88\ \text{pF}$):

$$\omega_u = \frac{g_{m1}}{C_C} = \frac{63.37\ \mu\text{S}}{1.88\ \text{pF}} \approx 33.71\ \text{Mrad/s}$$

$$t_{rise} \approx \frac{2.2}{\omega_u} = \frac{2.2}{33.71\ \text{Mrad/s}} \approx 65.26\ \text{ns}$$

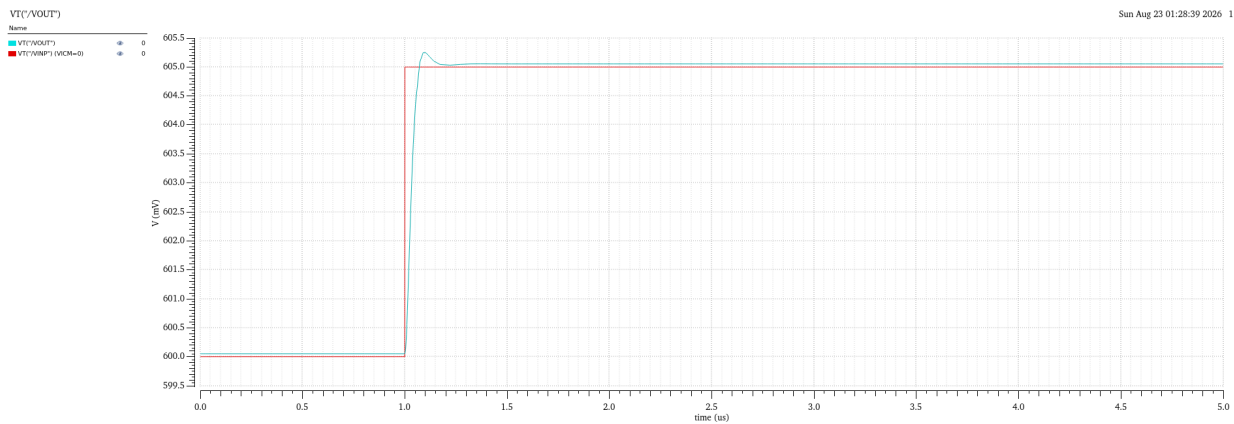


Figure 26. Small-signal transient step response (600mV to 605mV). The output exhibits a very fast rise time with minor, expected ringing.

Test	Output	Nominal	Spec	Weight	Pass/Fail
Transient_Analysis	VT(*/VOUT*)				
Transient_Analysis	VT(*/VINP*)				
Transient_Analysis	riseTime(VT(*/VOUT*) 0.6 n 0.6...	46.92n			

Figure 27. Cadence calculator extraction of the 10% to 90% rise time, yielding 46.92 ns.

Parameter	Hand Calculation	Simulation Result
Rise Time (t_{rise})	65.26 ns	46.92 ns

Table 7. Comparison of the first-order hand calculation versus the simulated second-order rise time.

Question

Do you see any ringing? Why?

Answer:

Yes, there is slight ringing visible. This happens because the OTA is not a perfect first-order system. The first-order hand calculation ($t_{rise} = 2.2\tau$) assumes a single dominant pole with 90° of phase margin. However, the actual OTA operates as a second-order system due to the non-dominant pole (ω_{p2}) and the LHP zero. Because the design targeted a phase margin near 70° , the system is slightly underdamped. This underdamped characteristic results in a simulation that is actually *faster* than the first-order theoretical model (46.92 ns vs. 65.26 ns), but at the cost of a small amount of expected ringing.

Part 5 (Optional): DC Closed-Loop AC Open-Loop OTA Simulation

Lab Requirement

Note that there will always be residual offset voltage between the first and second stages. This offset will drive the second stage output to one of the rails; thus, the biasing of the output stage will be disturbed. In order to avoid this problem, the DC OP point must be set by feedback. Use switches to connect the feedback loop in DC and break it in AC.

Implementation & Setup Strategy:

In standard open-loop AC analysis (as performed in Part 3), any residual systematic offset gets multiplied by the OTA's massive open-loop gain, which can push the DC output voltage away from $VDD/2$ and potentially drive the output transistors out of saturation.

To resolve this without altering the core OTA design, a dynamic testbench was constructed using ideal switches (`sp1tswitch`) to decouple the DC biasing phase from the AC small-signal phase:

- **DC State (Closed-Loop):** Switches S0 and S1 are closed, while S2 and S3 are open. This configures the OTA as a unity-gain buffer, forcing the DC output to perfectly balance at V_{ICM} (600.1 mV) and locking all transistors cleanly into saturation.
- **AC State (Open-Loop):** The instant the small-signal simulation begins, S0 and S1 open to break the feedback loop, while S2 and S3 close to inject the differential AC signal from the balun.

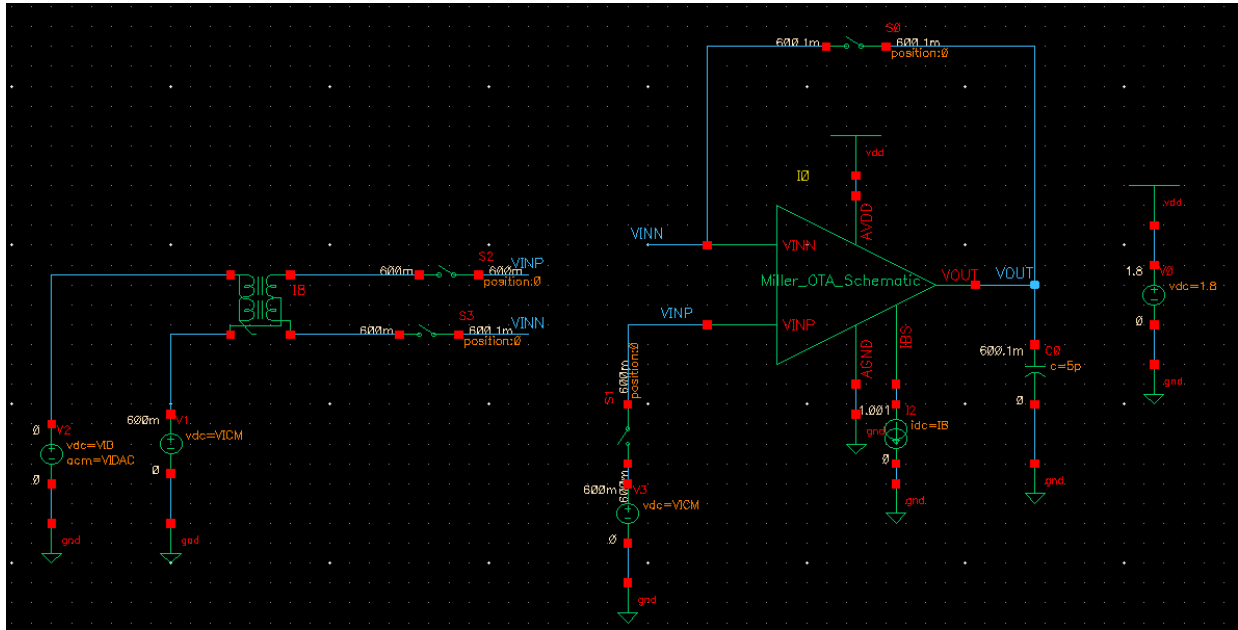


Figure 28. Annotated schematic of the DC Closed-Loop / AC Open-Loop testbench. The DC analysis confirms the output safely resting at 600.1 mV due to the active DC feedback path through S0.

By using this configuration, we can accurately simulate and extract the true open-loop characteristics without being hindered by DC offset limitations. As shown in the AC analysis below, the OTA maintains its high DC gain (76.04 dB). Furthermore, by probing the unity-gain crossover frequency (4.91 MHz), the phase is measured at -110° . Because the non-inverting open-loop phase natively starts at 0° , a phase of -110° mathematically equates to a Phase Margin of exactly 70° ($180^\circ - 110^\circ$), perfectly verifying the closed-loop STB stability measurements.

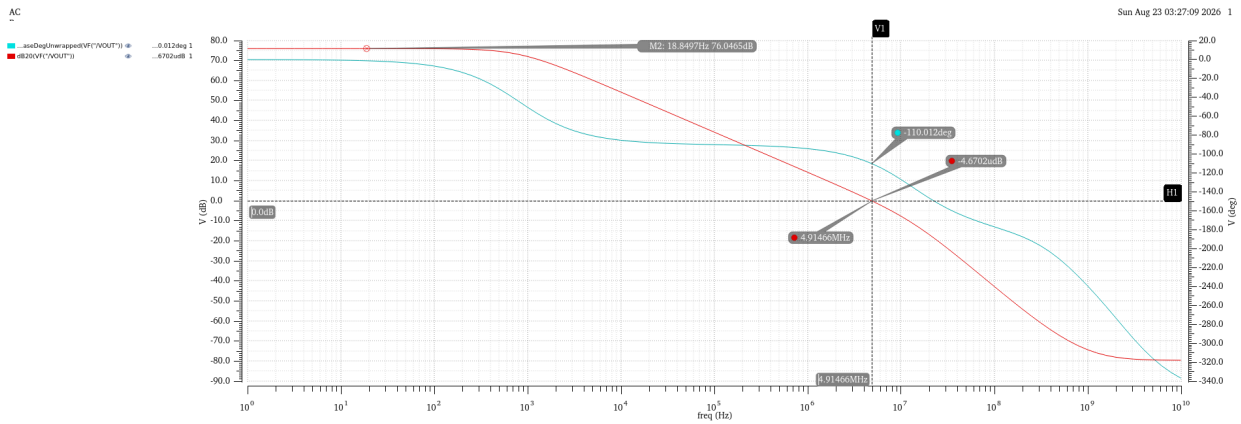


Figure 29. True Open-Loop AC Code Plot. The markers explicitly confirm the 76 dB DC gain and the flawless 70° Phase Margin at the unity-gain frequency.

Bonus: Tracking Lead Compensation

Lab Requirement

Implement the nulling resistor (R_z) using a transistor based on the Johns and Martin methodology to make compensation independent of process and temperature.

Implementation & Theory:

Using an ideal fixed resistor for lead compensation is impractical in physical layout. Furthermore, as temperature fluctuates, the transconductance of the second-stage driving transistor (g_{m7}) drifts. If R_z remains fixed while g_{m7} changes, the LHP zero will no longer perfectly cancel the non-dominant pole, degrading the carefully tuned phase margin over varying PVT conditions.

Following the Johns and Martin tracking lead compensation methodology, the ideal resistor was replaced with an NMOS transistor (M11) operating in the deep triode region. To ensure the zero consistently tracks the non-dominant pole, the resistance R_z must track the inverse transconductance of the second stage ($1/g_{m7}$). Therefore, our design goal is to force the product $R_z g_{m7} = 1$.

To achieve this dynamically, a dedicated bias network (M8, M9, M10) was constructed to generate a tracking gate voltage for M11. Johns and Martin prove that by matching the effective gate-source voltages of the bias network to the second stage, the product $R_z g_{m7}$ becomes purely dependent on transistor geometry (Equation 6.72):

$$R_z g_{m7} = \frac{(W/L)_{M7}}{(W/L)_{M11}} \sqrt{\frac{(W/L)_{M9}}{(W/L)_{M10}}}$$

Using this equation, the compensation network was mathematically sized by hand as follows:

- **Step 1: Sizing the Bias Network (M9 and M10)**

A 10 μA bias branch was created via PMOS M8. To match the effective voltages, Johns and Martin (Eq. 6.68) state that the W/L ratio must equal the current ratio between the second stage (M7) and the bias branch (M10):

$$\frac{I_{D(M7)}}{I_{D(M10)}} = \frac{40 \mu\text{A}}{10 \mu\text{A}} = 4 \implies (W/L)_{M10} = \frac{1}{4}(W/L)_{M7}$$

Given M7 has a width of 12.76 μm , the bottom diode M10 was sized to $W = 3.19 \mu\text{m}$ ($L = 520 \text{ nm}$). To form the tracking stack, the top diode M9 was sized identically to M10 ($W = 3.19 \mu\text{m}$).

- **Step 2: Sizing the Triode Resistor (M11)**

Because M9 and M10 are geometrically identical, the square root term in the master equation cancels out ($\sqrt{1} = 1$). To place the zero at infinity ($R_z = 1/g_{m7}$), the remaining geometry must equal 1:

$$1 = \frac{(W/L)_{M7}}{(W/L)_{M11}} \times 1 \implies (W/L)_{M11} = (W/L)_{M7}$$

Thus, the triode resistor M11 was sized identically to the second-stage driver M7: $W = 12.76 \mu\text{m}$ and $L = 520 \text{ nm}$.

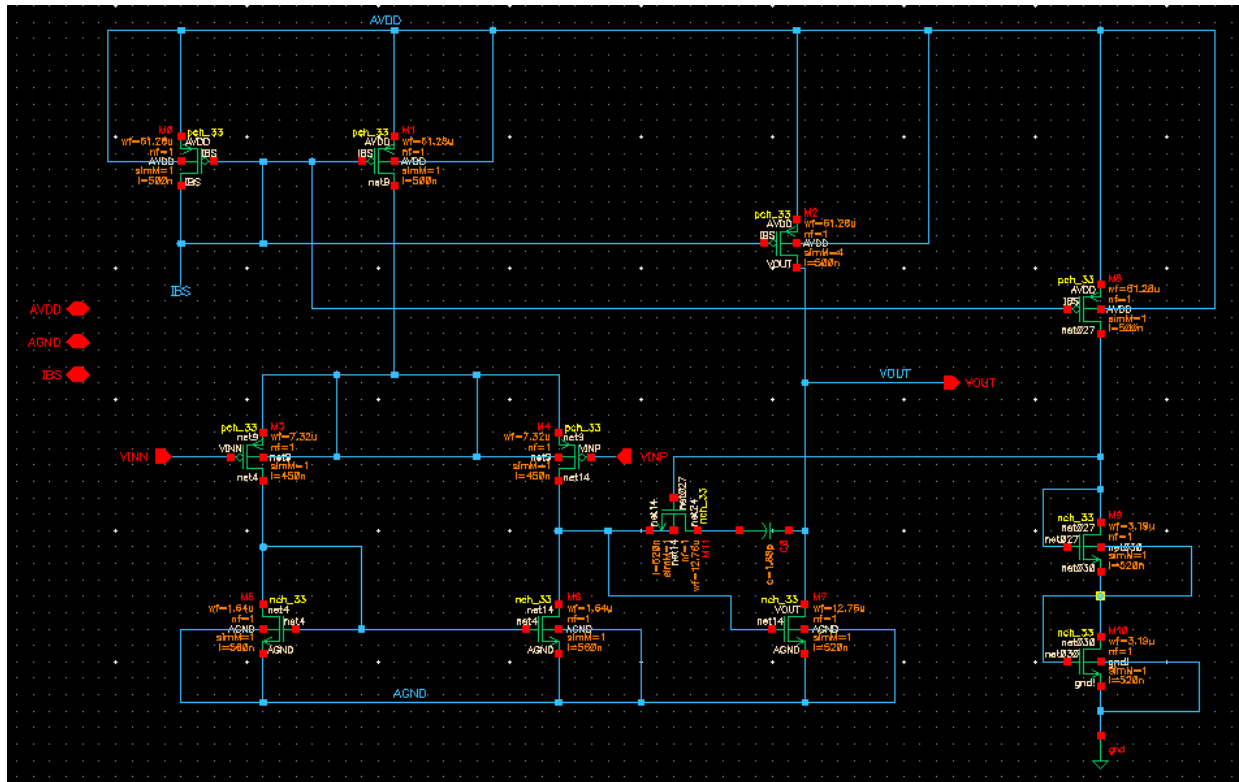


Figure 30. Implementation of the tracking lead compensation. The gate of the triode resistor M11 is dynamically biased by the M9/M10 stack.

Nominal Verification:

Running the STB analysis at nominal conditions (27°C) confirms the tracking circuit successfully replicates the ideal resistor's behavior, locking in a near-perfect target phase margin of 70.4° also using the SaveOP file I got gds so $1/g_{ds} = r_{ds}$ (which is our new R_z) got me 2.203k ohms close enough to the idealized hand calculation of mine at the very beginning (2.27k ohms).

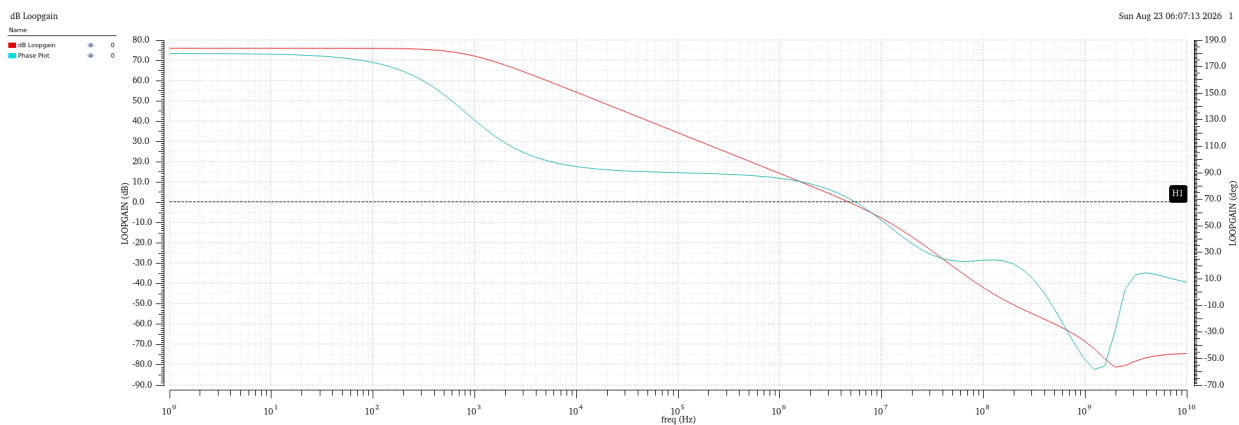


Figure 31. STB loop gain and phase response with the tracking triode resistor.

Test	Output	Nominal	Spec	Weight	Pass/Fail
DC_Analysis	none				
STB_Analysis	dB Loopgain				
STB_Analysis	Phase Plot				
STB_Analysis	Mag Loop gain				
STB_Analysis	GBW	5.229M			
STB_Analysis	UGF	4.913M			
STB_Analysis	Magnitude loop gain	6.305k			
STB_Analysis	Phase Margin	-109.6			

Figure 32. Extracted STB parameters showing a 4.91 MHz UGF and -109.6° phase (70.4° PM).

Temperature Tracking Robustness (The Proof):

To definitively prove the advantage of this architecture, a parametric temperature sweep from -40°C to 125°C was performed, comparing the new tracking circuit against the previous ideal resistor model.

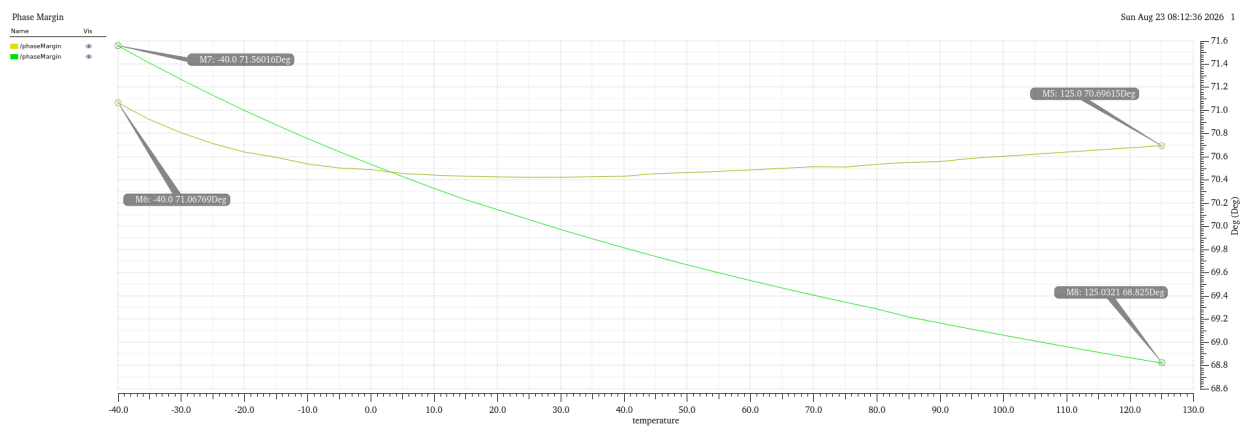


Figure 33. Phase Margin vs. Temperature. The uncompensated ideal resistor (green line) drifts noticeably as the transconductances change with temperature. Conversely, the Johns and Martin tracking compensation (yellow line) dynamically adjusts its resistance, keeping the phase margin flat and robust across the entire industrial temperature range.

The End of Mini Project 1 - Thank you
